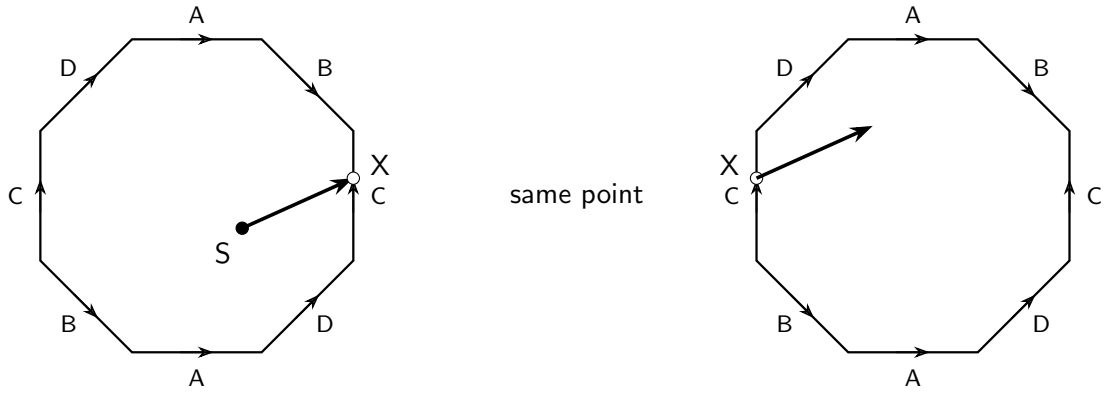
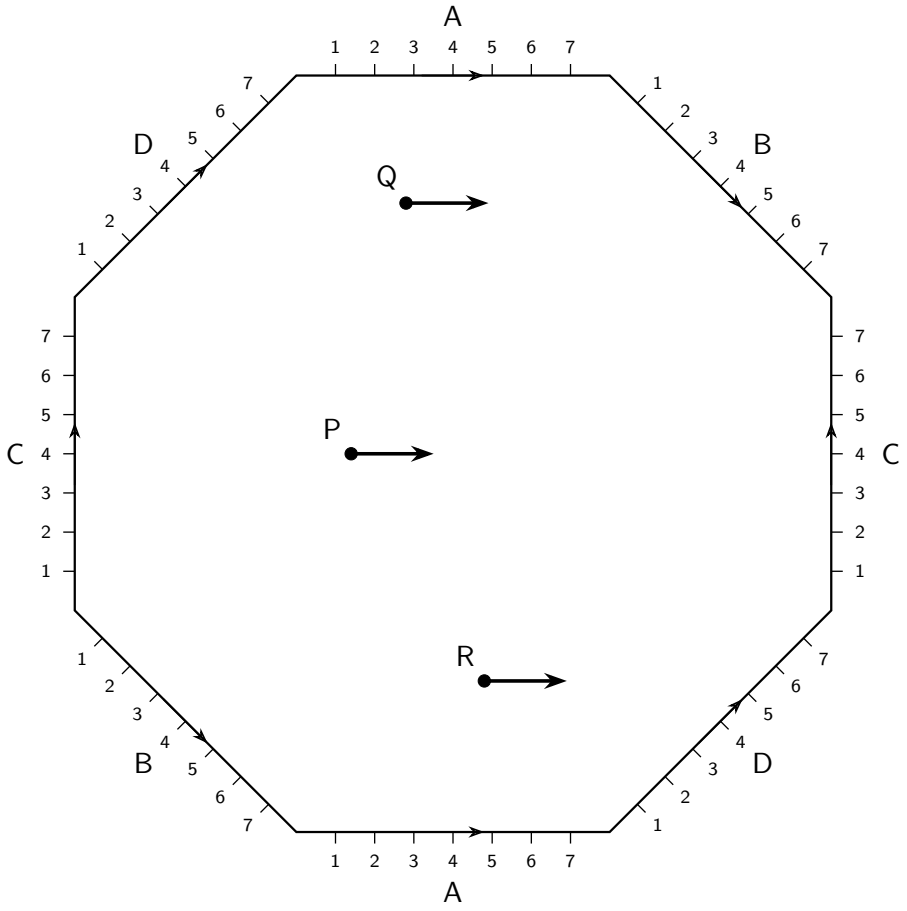
Slide same-letter edges together with their small arrows pointing the same way. At an edge, move to the same place on the other same-letter edge (match the numbered ticks) and keep the same travel direction. Stop at a corner. A trip closes when its starting point and direction both return.
One partner draws. The other checks each transfer with tracing paper and holds a separate arrow in the travel direction. Swap roles after each trip.

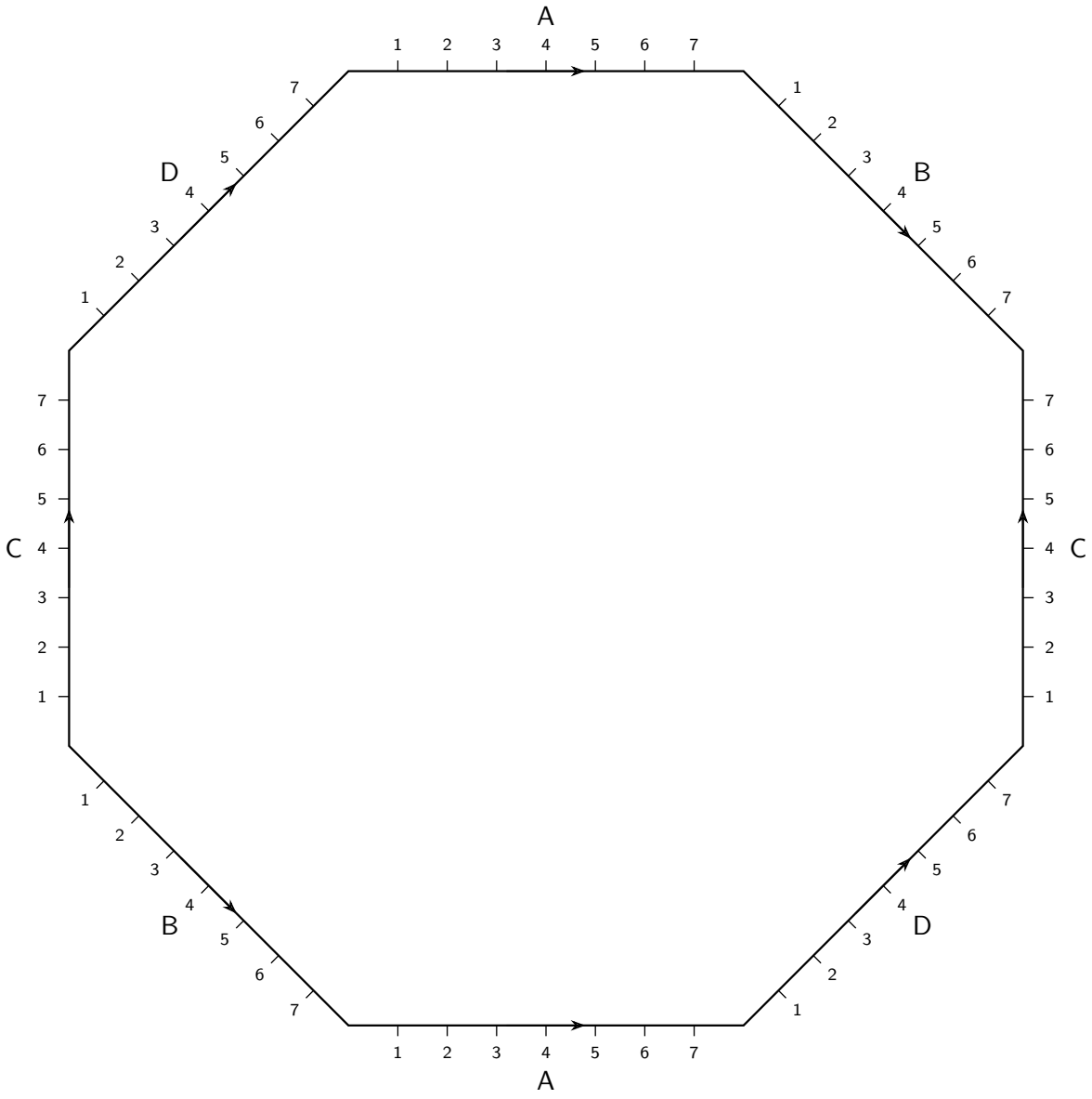


Start at S. Reach X. Resume at the matching X. Heavy arrows show travel.

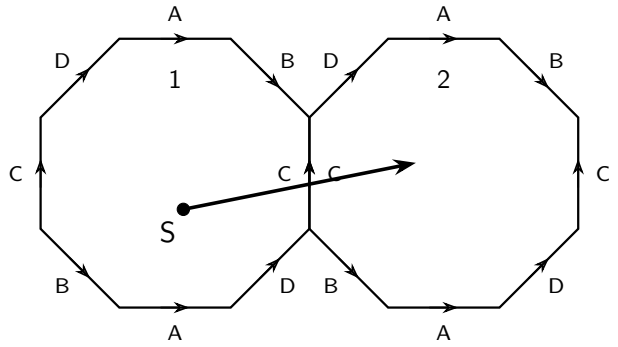
Problem 1: Follow each marked path until it closes or reaches a corner. Draw the three trips in different colors.

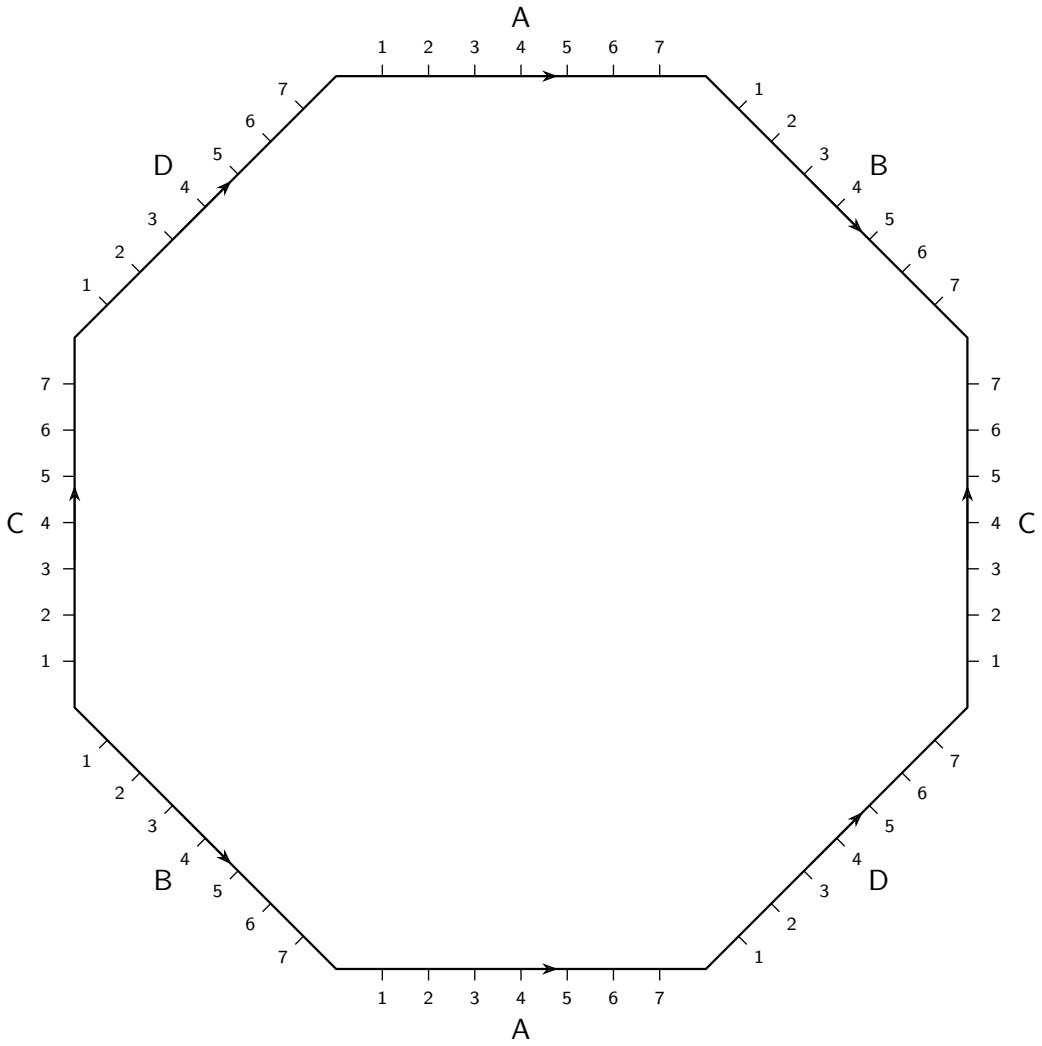


Problem 2: Choose starting dots inside the octagon and head to the right. Find all the different lengths of a closed trip that you can. Mark the starting places that give each length, and the starting places whose paths hit a corner. Count only the lines inside the octagon when you measure a trip.

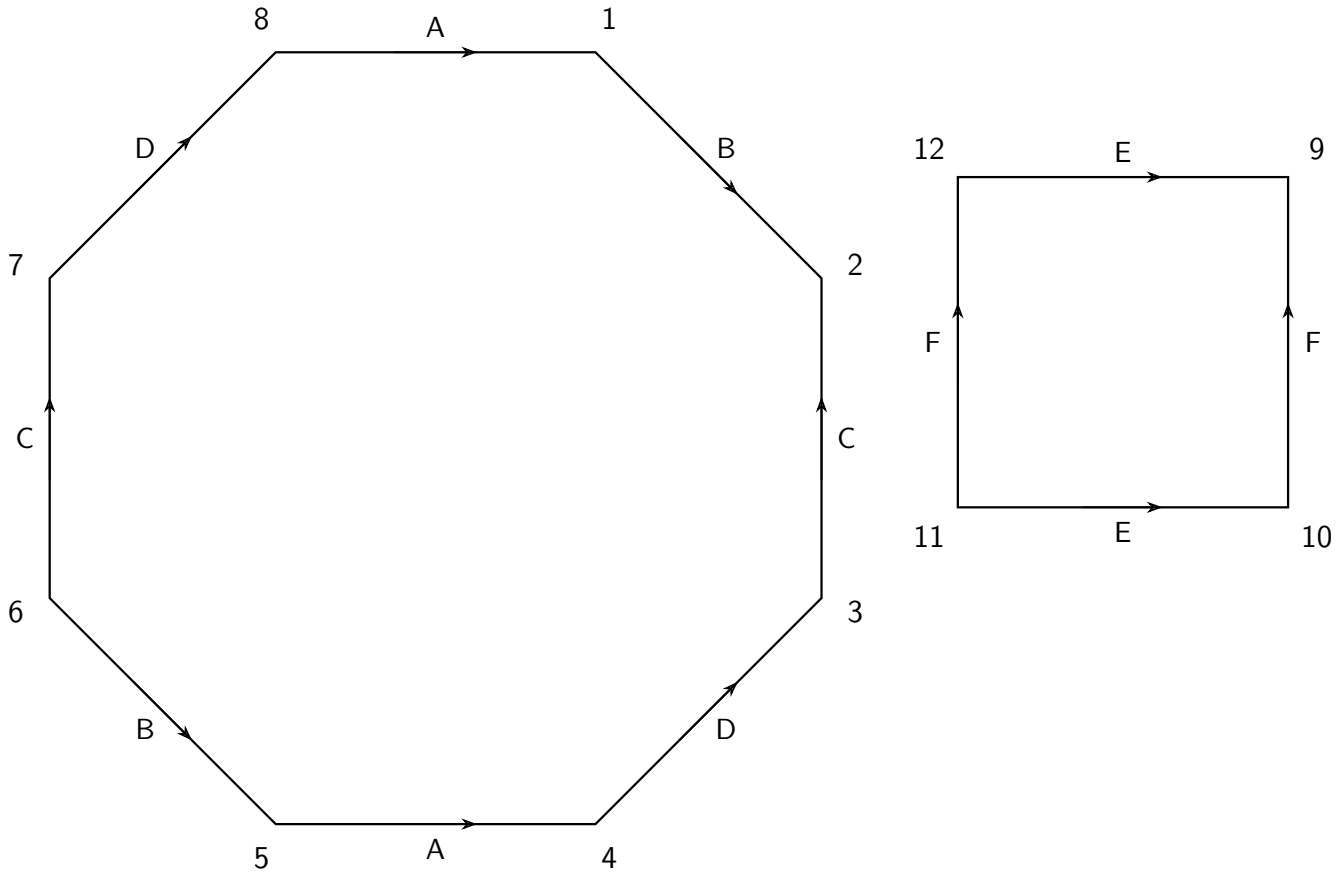


Problem 3: Start on an edge, away from a corner. Find a right-going closed trip whose two edge-to-edge stretches in the octagon have equal lengths. Show one full trip as a straight line across numbered tracings. To continue into a new tracing, slide its matching edge onto the edge you are leaving. Keep it facing the same way.

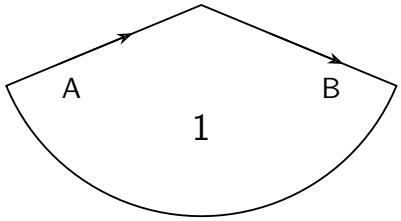


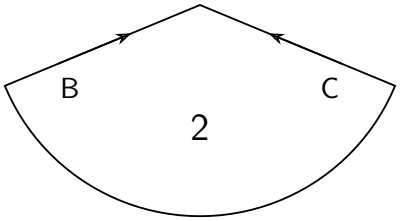


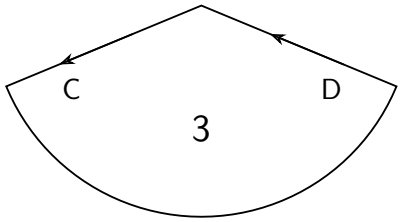
Problem 4: Match same-letter edges with their arrows pointing the same way. Which numbered corners become the same point? Put corners that meet in one group. Find all the groups for the octagon and for the square.

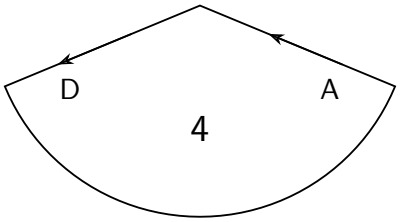


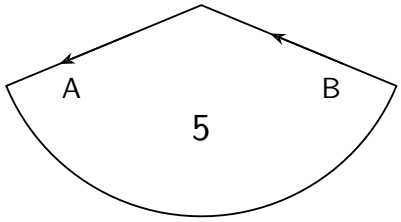
Problem 5: Cut out the corner pieces. Put the octagon corners around the point where they meet, in edge-matching order. Do the same with the square corners. How many full turns belong to each point? Let the pieces overlap as needed, keeping track of every numbered piece.

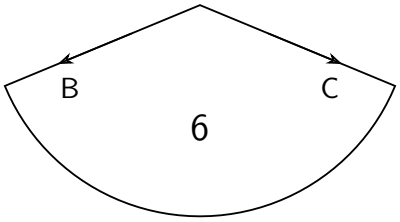


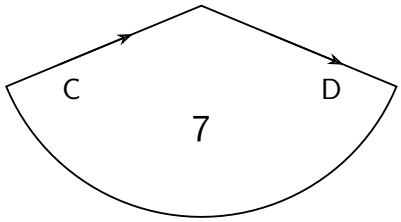


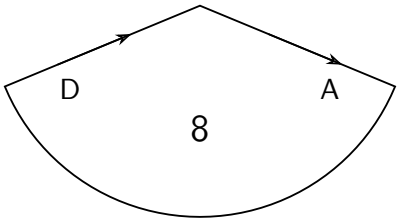


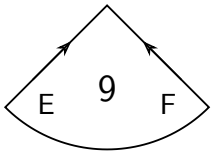


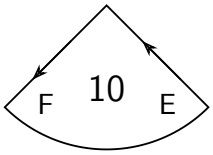


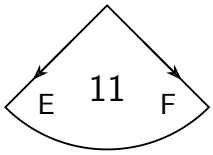


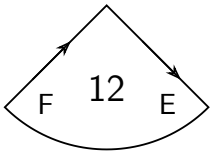


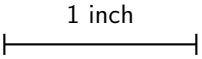








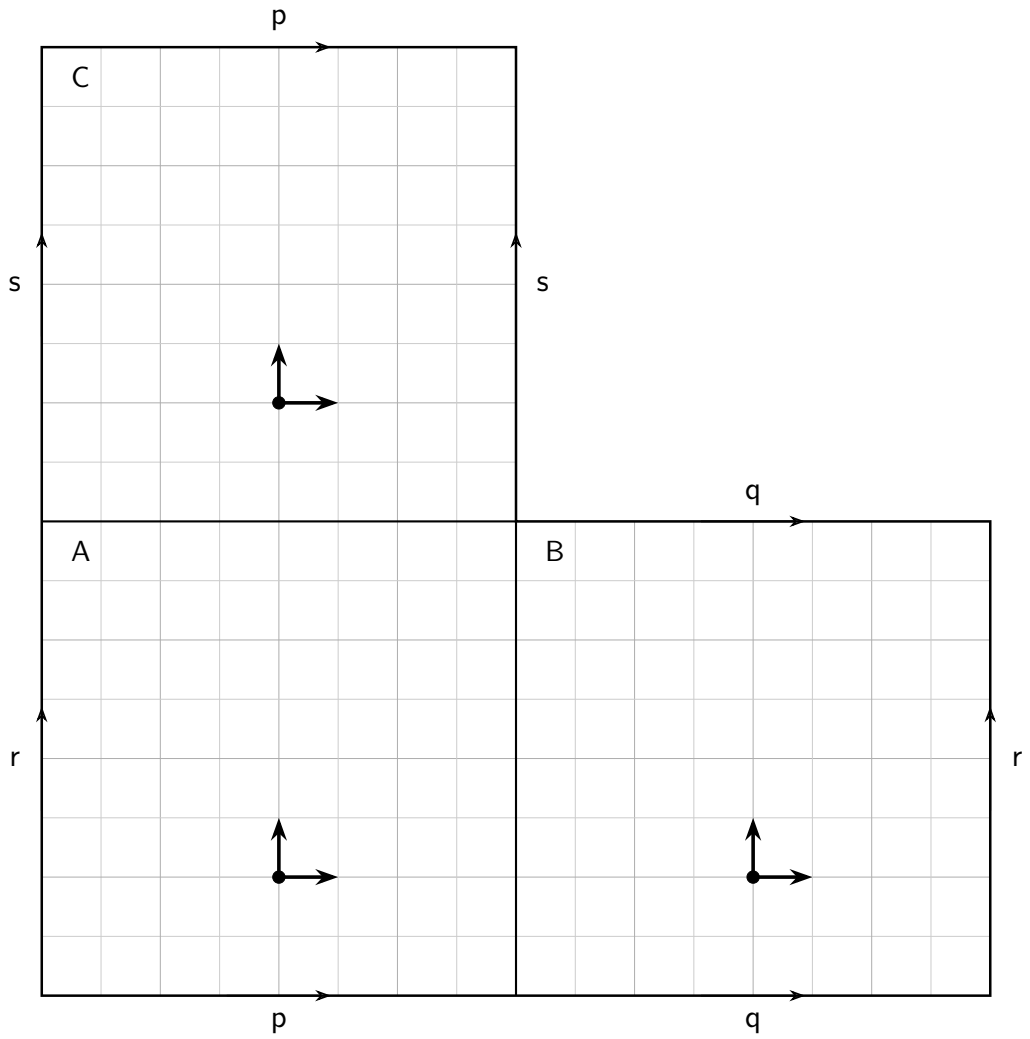




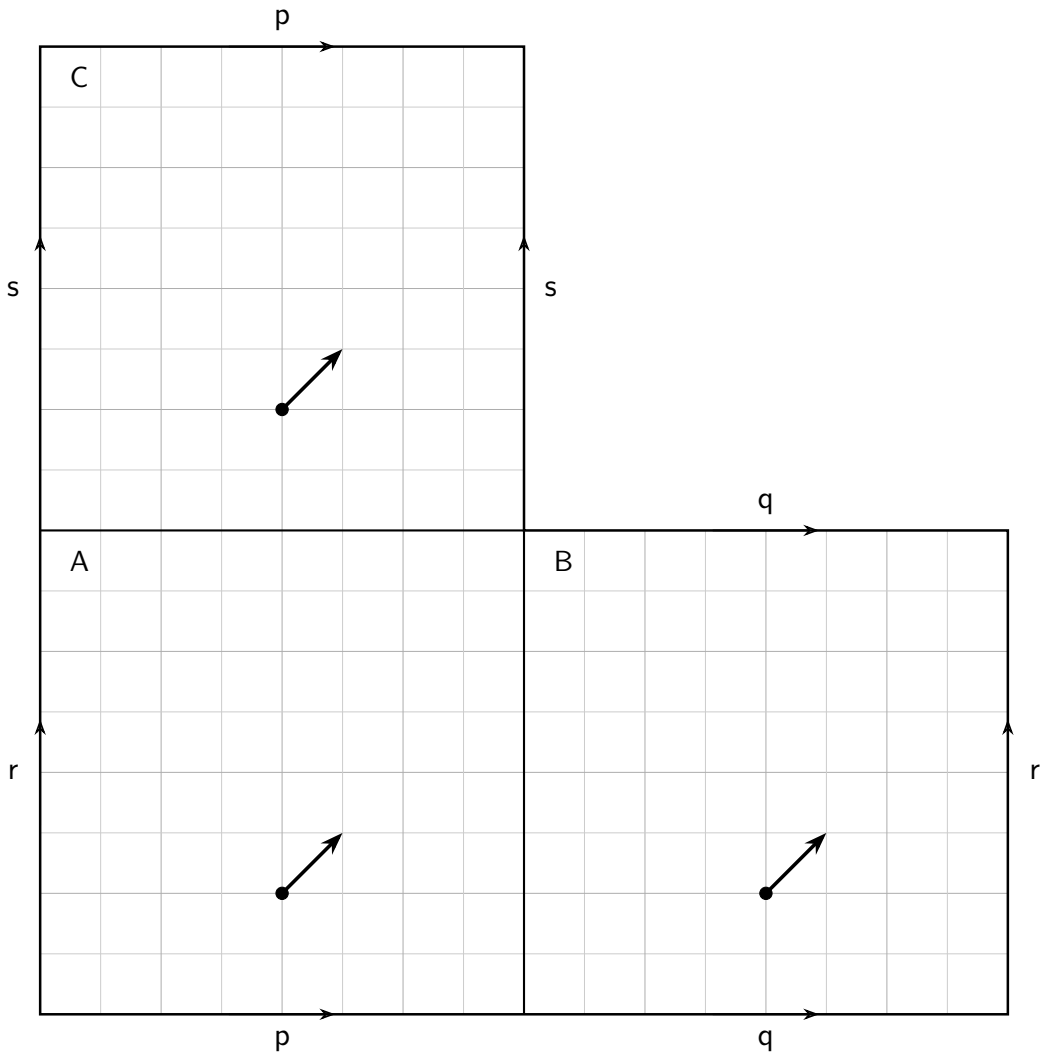
Octagon: _____ full turns Square: _____ full turns

Cross a shared side into the next square. The outside edges match by letters and arrows. Stop at any square corner.

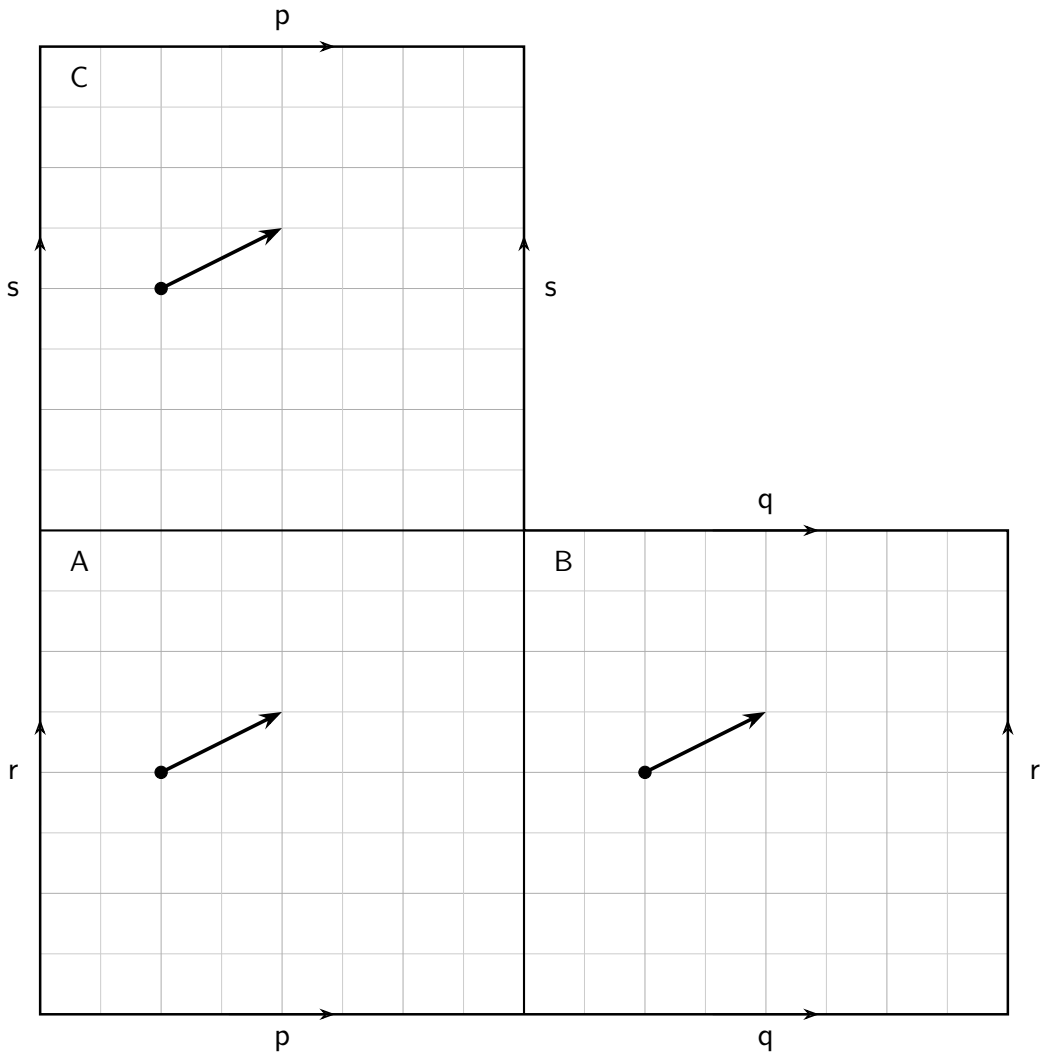
Problem 6: From each dot, make a closed trip to the right and a closed trip straight up. Which trips have the same length?



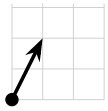
Problem 7: Find the first closed trip from each dot in the marked direction. Which dots does each trip pass, and in what order, before it returns to its starting dot?



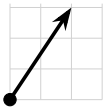
Problem 8: Find the first closed trip from each dot in the marked direction. Which starting squares give the shortest trip? How many times as long are the other trips?



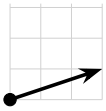
Problem 9: Try each of these straight directions from S. Each arrow gives a number of grid squares right and up. Which paths close, and which reach a corner? Must every straight direction with positive whole-number right and up amounts eventually close or reach a corner? Explain your answer.



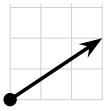
1 right, 2 up



2 right, 3 up



3 right, 1 up



3 right, 2 up

